

Pascal Jutras-Dubé

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SKILLS

Current Interests: Generative Modeling & Sampling, Diffusion LLMs, Stochastic Processes, Monte Carlo, Bayesian Inference, Optimization

Programming: Python, CUDA

Frameworks & Tools: JAX, PyTorch, Hydra, Weights & Biases, Git

EDUCATION

PhD	Computer Science	Purdue University	2023 – Present	GPA 3.95/4
MSc	Computer Science	University of Montreal	2020 – 2022	GPA 4.3/4.3
BSc	Mathematics and Computer Science	University of Montreal	2017 – 2020	

EXPERIENCE

Research Assistant

Jan 2023 - Present

Purdue University

- Designed Sticky Jump Diffusions unifying masked, continuous, and hybrid diffusion; introduced Denoising Hazard Matching to learn reverse dynamics from a single classifier, validated on text, image, and reasoning benchmarks.
- Designed a self-distilled one-step diffusion sampler that achieves competitive sample quality and statistical estimation with $1/256$ of the cost.

Teaching Assistant

Jan 2023 - Present

Purdue University, Graduate Statistical Machine Learning

- Wrote and graded assignments/exams, held office hours, and delivered lectures on diffusion models.

Research Intern

Jun 2021 - Nov 2022

National Bank of Canada

- Designed a scoring framework (linkability, information gain) of privacy-preserving data release strategies.

Software Engineer Intern

May 2019 - Aug 2019

GRAITEC

- Full-stack development of a CAD application for woodwork workflows; shipped features and fixes in an agile team environment.

PUBLICATIONS

Jutras-Dubé, P., Pynadath, P., Lu, J., Goa, Y., & Zhang, R.

Sticky Jump Diffusions: A Unifying Framework for Discrete, Continuous, and Hybrid Diffusion. Under Review at NeurIPS, 2026

Lee, S., **Jutras-Dubé, P.**, When, B., Zhang, R.

Neural Discrete Controlled Monte Carlo Samplers. ProbML Workshop @ ICML, 2026

Jutras-Dubé, P., Zhang, J., Wang, Z., & Zhang, R.

One-Step Diffusion Samplers via Self-Distillation and Deterministic Flow. AISTATS, 2026

Punyamoorthy, P.*, **Jutras-Dubé, P.***, Zhang, R., Aggarwal, V., Conover, D., & Bera, A.

Dynamic Obstacle Avoidance through Uncertainty-Based Adaptive Planning with Diffusion. IROS, 2025

Jutras-Dubé, P., Pynadath, P., & Zhang, R.

Single-Step Consistent Diffusion Samplers. FPI Workshop @ ICLR, 2025

Mesana, P., **Jutras-Dubé, P.**, Crowe, J., Vial, G., & Caporossi, G., & Gambs, S.

Measuring privacy/utility tradeoffs of format-preserving strategies for data release. Journal of Business Analytics, 2025

Jutras-Dubé, P., Al-Khasawneh, M. B., Yang, Z., Bas, J., Bastin, F., & Cirillo, C.

Copula-based transferable models for synthetic population generation. TR-C, 2024

Jutras-Dubé, P., Zhang, R., & Bera, A.

Adaptive planning with generative models under uncertainty. IROS, 2024

Mesana, P., **Jutras-Dubé, P.**, Crowe, J., Vial, G., & Caporossi, G.

Evaluating the risk of re-identification in data release strategies: An attacker-centric approach. HICSS, 2024

ADDITIONAL INFORMATION

Reviewer: NeurIPS 2026, ICML 2026, AISTATS 2026; ICLR 2025–2026; ICRA 2025; RA-L 2025–2026

Languages: French (native), English (proficient)

Almost Surely: My fashion startup for the STEM community: almost-surely.com

Awards: UdeM Graduate Scholarship (2021); DIRO Excellence (4×, 2020–2022); Fin-ML CREATE (2020)